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Chapter 15

Introduction to WANs

In this chapter, you are tested on the basics of WAN technologies and configuring a serial interface to connect to the WAN environment. You can expect some serial port configuration questions on the Cisco ICND1 and ICND2 exams, along with some common troubleshooting scenarios administrators are faced with.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Knowing WAN terminology
- Using different methods to connect to WANs
- Configuring serial links

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Know your different WAN technologies, such as circuit switched and packet switched. Circuit switch environments establish a pathway (a circuit) that is used for the duration of the communication. At the end of the conversation, the circuit is disconnected the next time communication occurs, a new circuit is established, and the pathway can be different. Know that the phone system and ISDN are examples of circuit switched environments.
- A dedicated leased line is a permanent point-to-point link that typically has more bandwidth than a circuit switched environment. T1 and T3 links are examples of leased lines.
- Packet switched environments deal with routing each packet a different way and then assembling them at the end. X.25 and frame relay are examples of packet switched networks.
- You can connect the serial port to an external CSU/DSU, which connects your network to a leased line such as that used by a T1 or T3 link. Routers may also have an internal, or integrated, CSU/DSU port, which connects directly to the leased line coming into your building from the service provider.

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Introduction to WAN Terminology

542–552 Choose the best answer(s).

542. What is the bandwidth of a T1 WAN link?

(A) 1.544 Mbps
(B) 64 Kbps
(C) 256 Kbps
(D) 128 Kbps

543. Which of the following represents a dedicated WAN link that you pay a monthly subscription fee for?

(A) Leased line
(B) Fast Ethernet
(C) Circuit switched
(D) Ethernet

544. Which WAN technology needs to first establish a connection between two points before sending the data and has all the data travel the same pathway?

(A) Leased line
(B) Packet switched
(C) Circuit switched
(D) Ethernet

545. What is the bandwidth of a BRI subscription?

(A) 1.544 Mbps
(B) 64 Kbps
(C) 256 Kbps
(D) 128 Kbps

546. What is the bandwidth of a T3 link?

(A) 1.544 Mbps
(B) 44.736 Mbps
(C) 256 Kbps
(D) 128 Kbps

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547. Which of the following technologies are dedicated leased lines?

(A) T1

(B) ISDN

(C) X.25

(D) Frame relay

548. What is the bandwidth of a PRI subscription?

(A) 1.544 Mbps

(B) 64 Kbps

(C) 256 Kbps

(D) 128 Kbps

549. Which type of WAN technology dedicates the bandwidth to the communication until the connection is terminated?

(A) Dedicated link

(B) Packet switch

(C) Circuit switch

(D) Leased line

550. Which of the following is considered a not-so-permanent WAN link?

(A) Leased line

(B) Fast Ethernet

(C) Circuit switched

(D) Ethernet

551. Which of the following is a circuit switching technology?

(A) T1

(B) ISDN

(C) X.25

(D) Frame relay

552. Which of the following WAN technologies are packet switching technologies? (Select all that apply.)

(A) T1

(B) ISDN

(C) X.25

(D) Frame relay

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Methods of Connecting to WANs

553–558 Choose the best answer(s).

553. Which device on a point-to-point link needs to have the clock rate set?

(A) The DTE device

(B) The first device connected

(C) The DCE device

(D) Both devices

554. Which of the following ports would you use to connect to a T1 connection?

(A) AUI

(B) F0/1

(C) Serial 0/0

(D) Console

555. What is the name of the DTE/DCE cable that is used to connect two routers together via their serial ports?

(A) Console cable

(B) Serial cable

(C) Back-to-back serial cable

(D) Crossover cable

556. You are troubleshooting your WAN connection. What layers of the OSI model are you troubleshooting? (Select two.)

(A) Application layer

(B) Physical layer

(C) Network layer

(D) Presentation layer

(E) Data link layer

557. Which of the following devices would typically connect the serial port on your router to a T1 line?

(A) External CSU/DSU

(B) Internal CSU/DSU

(C) External modem

(D) Internal modem

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558. Which of the following statements is true of a serial interface?

(A) The clock rate must be set on the DCE device.

(B) All Cisco routers are DCE devices.

(C) If the bandwidth is not set, you must set the clock rate.

(D) If the clock rate is not set, you must set the bandwidth.

Configuring Serial Links

559–579 Choose the best answer(s).

559. What is the default encapsulation protocol on a serial link for a Cisco router?

(A) PPP

(B) L2TP

(C) PPTP

(D) HDLC

560. You are connecting your Cisco router to a non-Cisco router over the serial link. Which of the following commands will you use?

(A) encapsulation hdlc

(B) encapsulation vpn

(C) encapsulation nat

(D) encapsulation ppp

561. You have disabled the serial 0/0 interface with the no shutdown command. How will the show interfaces command display the status of the interface?

(A) Serial 0/0 is administratively down, line protocol is down

(B) Serial 0/0 is down, line protocol is down

(C) Serial 0/0 is down, line protocol is up

(D) Serial 0/0 is up, line protocol is down

562. You have configured a test lab and you want to allow communication between two routers using PPP. Review the following code and identify any reasons why the two routers cannot communicate over the serial link.

```
MTL-R1(config)#interface s0/0
MTL-R1(config-if)#ip address 24.0.0.1 255.0.0.0
```

(A) The encapsulation protocol needs to be set.

(B) The IP address is incorrect.

(C) The subnet mask is incorrect.

(D) The name of the router needs to be changed.

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563. You are configuring the DCE end of a point-to-point serial link. What command would you type that is not typed on the DTE device?

(A) `encapsulation hdlc`

(B) `clock rate 64000`

(C) `encapsulation ppp`

(D) `bandwidth 64000`

564. Looking at the output below, what is the protocol configured on the serial port?

```
Router>show interfaces
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0009.7c82.4401 (bia 0009.7c82.4401)
  Internet address is 10.0.0.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never

 <remaining output omitted>

Serial1/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 10.1.0.1/24
  MTU 120 bytes, BW 128 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair

 <remaining output omitted>
```

(A) ARPA

(B) PPTP

(C) HDLC

(D) PPP

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565. Looking at the output shown below, what type of device is router R1?

```
R1>show controllers s1/0
Interface Serial1/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 64000
idb at 0x81081AC4, driver data structure at 0x81084AC0
SCC Registers:
General [GSMR]=0x2:0x00000000, Protocol-specific [PSMR]=0x8
Events [SCCE]=0x0000, Mask [SCCM]=0x0000, Status [SCCS]=0x00
Transmit on Demand [TODR]=0x0, Data Sync [DSR]=0x7E7E
Interrupt Registers:
Config [CICR]=0x00367F80, Pending [CIPR]=0x0000C000
Mask   [CIMR]=0x00200000, In-srv  [CISR]=0x00000000
Command register [CR]=0x580
```

(A) DTE

(B) SIS

(C) ABR

(D) DCE

566. Which of the following encapsulation protocols will allow communication with non-Cisco devices?

(A) PPP

(B) HDLC

(C) Frame relay

(D) IPX

567. Which packet switched technology uses a fixed length packet size known as a cell?

(A) ATM

(B) X.25

(C) Frame relay

(D) ISDN

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568. If you want to load a Cisco proprietary encapsulation protocol over the serial link, what command do you use?

(A) `encapsulation ppp`

(B) `encapsulation hdlc`

(C) `serial link ppp`

(D) `serial link hdlc`

569. Your router is unable to communicate over the WAN link that is connected to serial 0/0. What commands would you use to verify the status of the port? (Select two.)

(A) `show ip route`

(B) `show interfaces`

(C) `show ppp`

(D) `show hdlc`

(E) `show ip interface brief`

570. What are the requirements to configure PPP authentication between NY-R1 and TOR-R1? (Choose three.)

(A) Set the hostname on each router to the same name.

(B) Set the hostname on each router to a unique name.

(C) Create a username on each router named PPP.

(D) Create a username on each router that matches the hostname of the other router.

(E) Enable PPP authentication on both routers and specify either PAP or CHAP as the authentication protocol.

(F) Enable PPP authentication on the first router configured and specify either PAP or CHAP as the authentication protocol.

571. What command would you use to verify if a serial port is a DCE or DTE device?

(A) `show interface serial 0/0`

(B) `show ip interface`

(C) `show ip interface brief`

(D) `show controllers serial 0/0`

572. Your manager has asked you about the HDLC protocol on Cisco devices. Which of the following statements are true? (Select two.)

(A) It is used to connect routers from different vendors.

(B) It is the default encapsulation protocol for serial links on Cisco routers.

(C) It is the default layer-2 protocol for Ethernet connections.

(D) It is the default layer-2 protocol for Fast Ethernet connections.

(E) It is a Cisco proprietary implementation.

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573. You have a branch office connected to your head office using a WAN link over the serial ports. You have recently upgraded the router and are unable to communicate across the WAN link. Using the output below, what should you do to get the link up and running?

```
Router#show interface s1/0
Serial1/0 is up, line protocol is down (disabled)
  Hardware is HD64570
  Internet address is 10.1.0.1/24
  MTU 120 bytes, BW 128 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation PPP, loopback not set, keepalive set (10 sec)
  LCP Closed
```

(A) Connect the cable to the serial port properly.

(B) Change the encapsulation protocol.

(C) Change the IP address to 10.1.0.1/8.

(D) Set the bandwidth to 256 Kbps.

574. What encapsulation protocol would you use on the serial port in order to support authentication?

(A) PPTP

(B) HDLC

(C) SSTP

(D) PPP

575. The router administrator is troubleshooting the WAN link and wondering what the meaning of line protocol up means. (Select two.)

(A) The link is up.

(B) The IP address information is correct.

(C) Communication over the layer-2 protocol is working.

(D) Routing table has been updated.

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576. You are the network administrator who manages a router in the Toronto office named TOR-R1 and the New York office name NY-R1. You would like to configure PPP as the encapsulation protocol over the WAN link with PPP authentication between the two routers. Which of the following commands represent the configuration on the Toronto router?

(A) Use the following commands:

```
TOR-R1(config)#username TOR-R1 password mypass
TOR-R1(config)#interface serial 0/0
TOR-R1(config-if)#encapsulation hdlc
TOR-R1(config-if)#ppp authentication chap
```

(B) Use the following commands:

```
TOR-R1(config)#username NY-R1 password mypass
TOR-R1(config)#interface serial 0/0
TOR-R1(config-if)#encapsulation ppp
TOR-R1(config-if)#ppp authentication chap
```

(C) Use the following commands:

```
TOR-R1(config)#interface serial 0/0
TOR-R1(config-if)#encapsulation ppp
TOR-R1(config-if)#ppp authentication pap
```

(D) Use the following commands:

```
TOR-R1(config)#username NY-R1 password mypass
TOR-R1(config)#interface serial 0/0
TOR-R1(config-if)#ppp authentication chap
```

577. Which authentication protocol with PPP is considered the most secure?

(A) HDLC

(B) PPP

(C) PAP

(D) CHAP

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578. Which of the following authentication protocols supported by PPP should be used?

(A) PAP

(B) PPTP

(C) PPP

(D) CHAP

579. Which of the following commands would you use on the serial interface to configure CHAP authentication?

(A) `ppp authentication chap`

(B) `chap authentication`

(C) `set serial chap`

(D) `authentication chap`